

Variant classification guidelines

Miranda Durkie
Consultant Clinical Scientist
Rare Disease Lead Scientist
North East and Yorkshire Genomic
Medicine Service

It's all about

you & me & him & her & them.

A brief history of UK variant guidelines....!

ACTTAG CGT
CGTACT ACG
TCAGTACCGTACTTAG CLIN
GTGTCACTACCGTACT GEN

for the Interpretation (UVs) in Clinical Molecular

Prepared and edited by Jennie Bell¹, Dr

1. West Midlands Regional Genetics Laboratory, Edgbaston, Birmingham, B15 2TG, United Kingdom
2. Dept of human genetics, Radboud University Nijmegen, Nijmegen, The Netherlands
3. Dept of clinical genetics, VU University Amsterdam, Amsterdam, The Netherlands
4. National Genetics Reference Laboratory, Manchester, M13 9JH, United Kingdom

Guidelines ratified by the UK Clinical Molecular Genetic Laboratory Specialists (Vereniging, 2007).



VKGL

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rdmer⁴, Erik Sister-
Deans⁴ and Andrew

Foundation Trust, Mindel-
Northwick Park & St
ecastle-upon-Tyne, NE1
PO box 9101, 6500
straat 7, 1081 BT Amster-
ics, Saint Mary's Hospi-
Hospital, Crewe Road,
's Hospital, Cambridge,
f Laboratory Medicine,

Society (11th Janu-
Specialists (Vereni-
October, 2007).

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mber 2013).

Variant Classification Guidelines 2015



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ACMG STANDARDS AND GUIDELINES

Genetics
in Medicine

Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology

Sue Richards, PhD¹, Nazneen Aziz, PhD^{2,16}, Sherri Bale, PhD³, David Bick, MD⁴, Soma Das, PhD⁵, Julie Gastier-Foster, PhD^{6,7,8}, Wayne W. Grody, MD, PhD^{9,10,11}, Madhuri Hegde, PhD¹², Elaine Lyon, PhD¹³, Elaine Spector, PhD¹⁴, Karl Voelkerding, MD¹³ and Heidi L. Rehm, PhD¹⁵; on behalf of the ACMG Laboratory Quality Assurance Committee

	Benign		Pathogenic			
	Strong	Supporting	Supporting	Moderate	Strong	Very Strong
Population Data	MAF is too high for disorder (BS1/BS3) OR observation in controls inconsistent with disease penetrance (BS2)			Absent in population databases (PM2)	Prevalence in affecteds statistically increased over controls (PS4)	
Computational And Predictive Data		Multiple lines of computational evidence suggest no impact (BP4) Missense when only truncating cause disease (BP1) Silent variant with non-predicted splice impact (BP7) In-frame indels in repeat w/out known function (BP3)	Multiple lines of computational evidence support a deleterious effect on the gene /gene product (PP3)	Novel missense change at an amino acid residue where a different pathogenic missense change has been seen before (PM5) Protein length changing variant (PM6)	Same amino acid change as an established pathogenic variant (PS1)	Predicted null variant in a gene where LOF is a known mechanism of disease (PVS1)
Functional Data	Well-established functional studies show no deleterious effect (BP6)		Missense in gene with low rate of benign missense variants and path. missenses common (PP2)	Mutational hot spot or well studied functional domain without benign variation (PM1)	Well-established functional studies show a deleterious effect (PS3)	
Segregation Data	Non segregation with disease (BS4)		$N \leq 1/8$ if 1 family $N \leq 1/4$ if > 1 family	$N \leq 1/8$ if 1 family $N \leq 1/8$ if > 1 family	$N \leq 1/32$ if 1 family $N \leq 1/16$ if > 1 family	
De novo Data				De novo (without paternity & maternity confirmed) (PM8)	De novo (paternity & maternity confirmed) (PS2)	
Allelic Data		Observed in homo with a dominant variant (BP2) Observed in cis with a pathogenic variant (BP2)		For recessive disorders, detected in homo with a pathogenic variant (PM1)		
Other Database		Reputable source w/out shared data = benign (BP6)	Reputable source = pathogenic (PS5)			
Other Data		Found in case with an alternate cause (BP5)	Patient's phenotype or is highly specific for gene (PM4)			

Figure 1. ACMG-AMP Pathogenicity Classification Guidelines Updated to Include Cosegregation

The summary table of evidence categories and levels from the ACMG-AMP pathogenicity classification guideline paper by Richards et al.³ is shown and is updated to include the proposed cosegregation thresholds as outlined in Table 1. The strongest evidence level supported by a given N is selected. Figure adapted with permission from Richards et al.¹

Richards et al (2015)

Pathogenic	(i) 1 Very strong (PVS1) AND (a) ≥1 Strong (PS1-PS4) OR (b) ≥2 Moderate (PM1-PM6) OR (c) 1 Moderate (PM1-PM6) and 1 supporting (PP1-PP5) OR (d) ≥2 Supporting (PP1-PP5) (ii) ≥2 Strong (PS1-PS4) OR (iii) 1 Strong (PS1-PS4) AND (a) ≥3 Moderate (PM1-PM6) OR (b) 2 Moderate (PM1-PM6) AND ≥2 Supporting (PP1-PP5) OR (c) 1 Moderate (PM1-PM6) AND ≥4 supporting (PP1-PP5)
Likely pathogenic	(i) 1 Very strong (PVS1) AND 1 moderate (PM1-PM6) OR (ii) 1 Strong (PS1-PS4) AND 1-2 moderate (PM1-PM6) OR (iii) 1 Strong (PS1-PS4) AND ≥2 supporting (PP1-PP5) OR (iv) ≥3 Moderate (PM1-PM6) OR (v) 2 Moderate (PM1-PM6) AND ≥2 supporting (PP1-PP5) OR (vi) 1 Moderate (PM1-PM6) AND ≥4 supporting (PP1-PP5)
Benign	(i) 1 Stand-alone (BS1) OR (ii) ≥2 Strong (BS1-BS4)
Likely benign	(i) 1 Strong (BS1-BS4) and 1 supporting (BP1-BP7) OR (ii) ≥2 Supporting (BP1-BP7)
Uncertain significance	(i) Other criteria shown above are not met OR (ii) the criteria for benign and pathogenic are contradictory

Class	Definition	Probability
5	Pathogenic	>0.99
4	Likely Pathogenic	0.90-0.99
3	Uncertain	0.10-0.90
2	Likely benign	0.01-0.10
1	Benign	<0.01

Clinical Genome Resource (ClinGen) activity

SVI General Recommendations for Using ACMG/AMP Criteria

SVI provides general recommendations for using the ACMG/AMP criteria to improve consistency in usage and transparency in classification rationale.

- Guidance Classifying Variants in Genes Associated with Multiple Disorders
- Guidance to VCEPs regarding the use of gnomAD v4 (August 2024)
- Guidance on how to rename criteria codes when strength of evidence is modified
- BA1: Updated Recommendation for the ACMG/AMP Stand Alone Pathogenicity Criterion for Variant Classification
 - BA1 Exception List (July 2018)
 - BA1 Exception List Nomination Form
- PVS1: Recommendations for Interpreting the Loss of Function PVS1 ACMG/AMP Variant Criteria
 - ClinGen SVI PVS1 Decision Tree Editable
- PVS1, PS1, PP3, BP4, BP7: Application of the ACMG/AMP Framework to Capture Evidence Relevant to Predicted and Observed Impact on Splicing: Recommendations from the ClinGen SVI Splicing Subgroup
 - ClinGen SVI Splicing Subgroup feedback form
 - ClinGen SVI Splicing Subgroup - Response to feedback (March 2024)
- PS2/PM6: Recommendation for de novo PS2 and PM6 ACMG/AMP criteria (Version 1.1)
- PS3/BS3: Recommendations for application of the functional evidence PS3/BS3 criterion using the ACMG/AMP sequence variant interpretation framework
 - SVI Functional Assay Documentation Worksheet
- PM2: Recommendation for Absence/Rarity Criterion PM2 (Version 1.0)
- PM3: Recommendation for in trans Criterion PM3 (Version 1.0)
- PP1/BS4, PP4: ClinGen guidance for use of the PP1/BS4 co-segregation and PP4 phenotype specificity criteria for sequence variant pathogenicity classification
- PP3/BP4: Calibration of computational tools for missense variant pathogenicity classification and ClinGen recommendations for PP3/BP4
- PP5/BP6: Recommendation for reputable source PP5 and BP6 ACMG/AMP criteria



Recommendations supported by SVI

- Fitting a naturally scaled point system to the ACMG/AMP variant classification guidelines
- Recommendations for clinical interpretation of variants found in non-coding regions of the genome



CSpec Registry

The ClinGen Criteria Specification (CSpec) Registry UI displays Variant Curation Expert Panel (VCEP) Criteria Specifications approved by the SVI VCEP Review Committee.

- Criteria Specification Registry (CSpec)

ClinGen Variant Curation Expert Panel (VCEP) guidelines

<https://cspec.genome.network/cspec/ui/svi/>

Charcot-Marie-Tooth Disease VCEP

ClinGen Charcot-Marie-Tooth Disease Expert Panel Specifications to the ACMG/AMP Variant Interpretation Guidelines for SORD Version 1.0.0	SORD (HGNC:11184) 	Charcot-Marie-Tooth disease (MONDO:0015626)  Autosomal recessive inheritance	Pilot Rules In Prep	1.0.0	  
ClinGen Charcot-Marie-Tooth Disease Expert Panel Specifications to the ACMG/AMP Variant Interpretation Guidelines for MFN2 Version 1.0.0	MFN2 (HGNC:18872) 	obsolete axonal hereditary motor and sensory neuropathy (MONDO:0018772)  Autosomal dominant inheritance	Classification Rules In Prep	1.0.0	  
ClinGen Charcot-Marie-Tooth Disease Expert Panel Specifications to the ACMG/AMP Variant Interpretation Guidelines for GDAP1 Version 1.0.0	GDAP1 (HGNC:13958) 		Classification Rules In Prep	1.0.0	  

Coagulation Factor Deficiency VCEP

ClinGen Coagulation Factor Deficiency Expert Panel Specifications to the ACMG/AMP Variant Interpretation Guidelines for F8 Version 1.0.0	F8 (HGNC:3548) 	hemophilia A (MONDO:0010602)  X-linked inheritance	Released	1.0.0	  
ClinGen Coagulation Factor Deficiency Expert Panel Specifications to the ACMG/AMP Variant Interpretation Guidelines for F9 Version 1.0.0	F9 (HGNC:3551) 	hemophilia B (MONDO:0010604)  X-linked inheritance	Released	1.0.0	  
ClinGen Coagulation Factor Deficiency Expert Panel Specifications to the ACMG/AMP Variant Interpretation Guidelines for F7 Version 1.0.0	F7 (HGNC:3546) 	factor VII deficiency (MONDO:0002344)  Autosomal recessive inheritance	Classification Rules In Prep	1.0.0	  

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UK ACGS Variant Classification Guidelines

ACMG STANDARDS AND GUIDELINES

Genetics
in Medicine

Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology

Sue Richards, PhD¹, Nazreen Aziz, PhD^{1*}, Sheri Bale, PhD¹, David Bick, MD², Soma Das, PhD³, Julie Gastier-Foster, PhD^{1*}, Wayne H. Gandy, MD, PhD^{1*}, Madhuri Hegde, PhD¹, Elaine Lyns, PhD¹, Elaine Sperber, PhD¹, Karl Steinberg, MD^{1*} and Heidi L. Rehm, PhD¹
on behalf of the ACMG Laboratory Quality Assurance Committee



ACGS Best Practice Guidelines for Variant Classification in Rare Disease 2020

San Eilert^{1,2}, Emma L. Bagle^{1,2,3}, Alison Callaway², Ian Berry², Natalie Forrester², Clare Turnbull², Martina Owens², Diana M Eccles², Stephen Alden², Richard Scott^{1,2}, Zandra C Deans¹, Tracy Lester¹, Jo Campbell¹, William G Newman^{1,2}, Simon Ramsden² and Dominic J McMullan^{2*}



ACGS Best Practice Guidelines for Variant Classification in Rare Disease 2024

Winanda Darkie^{1*}, Emma-Jane Cassidy¹, Ian Berry², Martina Owens², Clare Turnbull², Richard H Scott¹, Robert W Taylor^{1,2}, Zandra C Deans¹, San Eilert², Emma L Bagle^{1,2,3} and Dominic J McMullan^{2*}
*contributed equally

2015

2017

2018

2019

2020

2020

2024

2024

ACMG TECHNICAL STANDARDS

Genetics
in Medicine

Technical standards for the interpretation and reporting of constitutional copy-number variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics (ACMG) and the Clinical Genome Resource (ClinGen)

Eryn Rooney Riggs, MS, CGC¹, Erica F. Andersen, PhD^{2,3}, Athena M. Cherry, PhD⁴, Sibel Kantani, PhD⁵, Hutton Kearney, PhD⁶, Anikita Patel, PhD¹, Gordana Raca, MD, PhD¹, Deborah L Ritter, PhD⁷, Sarah T. South, PhD^{1,8}, Erik C. Thorland, PhD⁹, Daniel Pinada-Alvarez, MD^{1,1}, Swarnon Aradhya, PhD^{1,11} and Christi Lee Martin, PhD¹



ACGS Best Practice Guidelines for Variant Classification in Rare Disease 2024

Winanda Darkie^{1*}, Emma-Jane Cassidy¹, Ian Berry², Martina Owens², Clare Turnbull², Richard H Scott¹, Robert W Taylor^{1,2}, Zandra C Deans¹, San Eilert², Emma L Bagle^{1,2,3} and Dominic J McMullan^{2*}
*contributed equally

Objectives for the ACGS guidelines

ACGS Best Practice Guidelines for Variant Classification in Rare Disease 2024

Miranda Durkie^{*1}, Emma-Jane Cassidy^{*2}, Ian Berry³, Martina Owens⁴, Clare Turnbull^{5,6}, Richard H Scott^{7,8}, Robert W Taylor^{9,10}, Zandra C Deans¹¹, Sian Ellard⁴, Emma L Baple^{4,12,13} and Dominic J McMullan¹⁴

* contributed equally

- Unify the interpretation of SNVs and CNVs in UK laboratories.
 - Provide an updated version of the SNV guidelines for use until the major revision of the ACMG v4.0 guidelines has been released, tested and is ready for implementation.
 - Align with ClinGen SVI and VCEP recommendations
 - Include guidelines for reporting of all variants, including risk alleles, noncoding, mitochondrial, reduced penetrance and hypomorphic sequence variants
- **Standardised, consistent, clinically relevant classifications across the NHS in ONE document**

Key features in ACGS guidelines

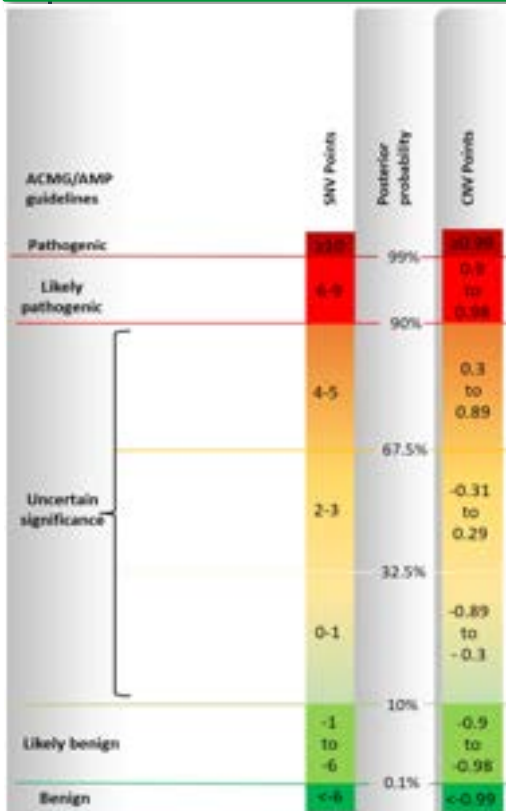


Figure 3: The point systems used to classify variants using SNV and CNV guidelines with differing levels of evidence in support of pathogenicity

Fully points based system

- Bayesian-derived **evidence points** (Tavtigian *et al.*, 2020) for SNVs and non-Bayesian points for CNVs – **combined into same figure**

TABLE 2 Point values for ACMG/AMP strength of evidence categories

Evidence Strength	Point scale	
	Pathogenic	Benign
Indeterminate	0	0 ^a
Supporting	1	-1
Moderate	2	-2 ^b
Strong	4	-4
Very strong	8	-8 ^b

- Sub-division of VUS and clear VUS reporting recommendations**



It is **essential** that **Clinical Scientists** use their **professional judgement** for variant classification. These guidelines have been developed to **augment the ACMG/AMP** framework for variant classification. They **cannot describe every scenario and the weighting** that should be applied for each evidence criterion. **Many variants** prioritised by a bioinformatic pipeline **will not require formal classification**. Clinical Scientists must identify those variants that require a full classification, then collate and **critically appraise** evidence for/against pathogenicity in the context of the individual case under consideration.

Cancer Variant Interpretation Group: CanVIG-UK

Cancer genetics

OPEN ACCESS

POSITION STATEMENT

Cancer Variant Interpretation Group UK (CanVIG-UK): an exemplar national subspecialty multidisciplinary network

Alice Garrett¹, Alison Callaway,^{2,3} Miranda Durkie,⁴ Conkut Cubuk,^{1,5}
Mary Alikian,¹ George J Burghel,⁶ Rachel Robinson,⁷ Louise Izatt,⁸
Sabrina Talukdar,⁹ Lucy Side,¹⁰ Treena Cranston,¹¹ Sheila Palmer-Smith,¹²
Diana Baralle,¹³ Ian R Berry,⁷ James Drummond,¹⁴ Andrew J Wallace,⁴
Gail Norbury,¹⁵ Diana M Eccles,¹⁶ Sian Ellard,¹⁸ Fiona Lalloo,⁵
D Gareth Evans,^{6,17} Emma Woodward,^{6,17} Marc Tischkowitz,¹⁸ Helen Hanson,^{1,9}
Clare Turnbull^{1,18} On behalf of the CanVIG-UK

2020

- Set up 2017
- Led by Professor Clare Turnbull
- Monthly webinars and monthly steering and advisory group meetings (CStAG)



Appendix 1: CanVIG-UK Consensus Specification for Cancer Genotyping
Version of ACOG Best Practice Guidelines for Variant Classification (V1.1) (2019)

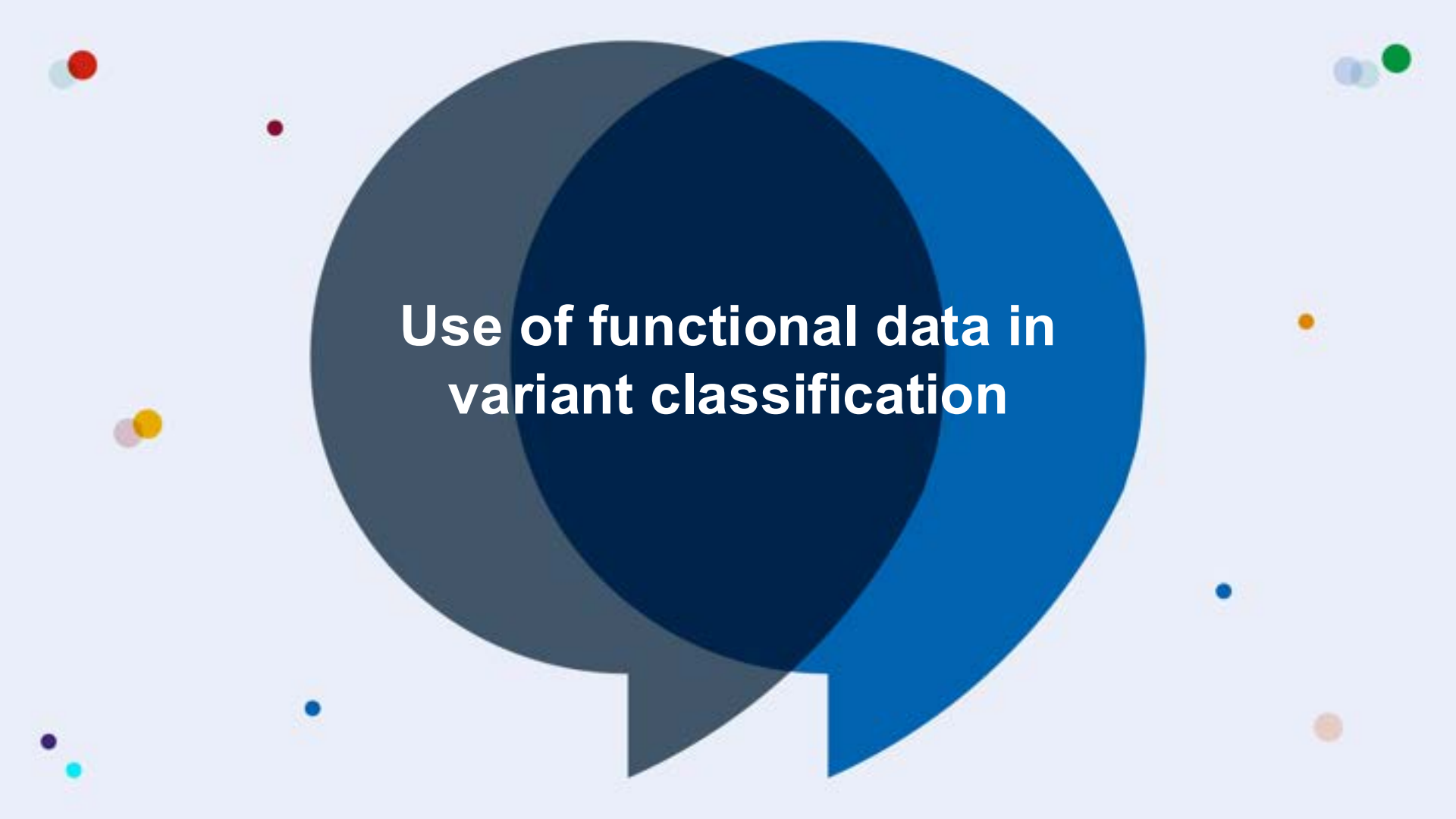
Guidelines:

- For the following evidence base, CanVIG-UK best practice specification for V1.1 is set out below. This document is a summary of the evidence base for the guidelines.
- For the following evidence base, CanVIG-UK best practice specification for V1.1 is set out below. This document is a summary of the evidence base for the guidelines.
- For the following evidence base, CanVIG-UK best practice specification for V1.1 is set out below. This document is a summary of the evidence base for the guidelines.

Table 1: Variant Classification

Variant	Pathogenic	Benign	Uncertain	Not Classifiable
BRCA1/2	Yes	No	Yes	No
TP53	Yes	No	Yes	No
MLH1	Yes	No	Yes	No
MSH2	Yes	No	Yes	No
PTEN	Yes	No	Yes	No
PIK3CA	Yes	No	Yes	No
NRAS	Yes	No	Yes	No
KRAS	Yes	No	Yes	No
EGFR	Yes	No	Yes	No
HER2	Yes	No	Yes	No
VEGFR	Yes	No	Yes	No
IGF1R	Yes	No	Yes	No
RET	Yes	No	Yes	No
RETN1	Yes	No	Yes	No
RETN2	Yes	No	Yes	No
RETN3	Yes	No	Yes	No
RETN4	Yes	No	Yes	No
RETN5	Yes	No	Yes	No
RETN6	Yes	No	Yes	No
RETN7	Yes	No	Yes	No
RETN8	Yes	No	Yes	No
RETN9	Yes	No	Yes	No
RETN10	Yes	No	Yes	No
RETN11	Yes	No	Yes	No
RETN12	Yes	No	Yes	No
RETN13	Yes	No	Yes	No
RETN14	Yes	No	Yes	No
RETN15	Yes	No	Yes	No
RETN16	Yes	No	Yes	No
RETN17	Yes	No	Yes	No
RETN18	Yes	No	Yes	No
RETN19	Yes	No	Yes	No
RETN20	Yes	No	Yes	No
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RETN25	Yes	No	Yes	No
RETN26	Yes	No	Yes	No
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RETN30	Yes	No	Yes	No
RETN31	Yes	No	Yes	No
RETN32	Yes	No	Yes	No
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RETN34	Yes	No	Yes	No
RETN35	Yes	No	Yes	No
RETN36	Yes	No	Yes	No
RETN37	Yes	No	Yes	No
RETN38	Yes	No	Yes	No
RETN39	Yes	No	Yes	No
RETN40	Yes	No	Yes	No
RETN41	Yes	No	Yes	No
RETN42	Yes	No	Yes	No
RETN43	Yes	No	Yes	No
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RETN46	Yes	No	Yes	No
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RETN90	Yes	No	Yes	No
RETN91	Yes	No	Yes	No
RETN92	Yes	No	Yes	No
RETN93	Yes	No	Yes	No
RETN94	Yes	No	Yes	No
RETN95	Yes	No	Yes	No
RETN96	Yes	No	Yes	No
RETN97	Yes	No	Yes	No
RETN98	Yes	No	Yes	No
RETN99	Yes	No	Yes	No
RETN100	Yes	No	Yes	No

We are now on 10th version of Consensus guidelines; have 14 gene-specific guidelines including **BRCA1/BRCA2** guidelines on 11th version

The background features two large, overlapping circles: a dark grey one on the left and a blue one on the right. The intersection of these circles is a darker blue. Scattered around the main circles are several small, semi-transparent dots in various colors including red, yellow, green, orange, and blue.

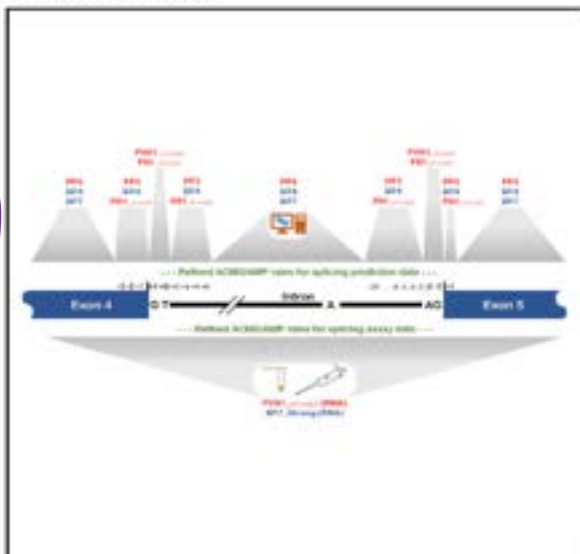
Use of functional data in variant classification

Use of RNA splicing data in variant classification

ARTICLE

Using the ACMG/AMP framework to capture evidence related to predicted and observed impact on splicing: Recommendations from the ClinGen SVI Splicing Subgroup

Graphical abstract



Authors

Logan C. Walker, Miguel de la Hoya, George A.R. Wiggins, ..., Steven M. Harrison, Amanda B. Spurdle, ClinGen Sequence Variant Interpretation Working Group

Correspondence

sharriso@broadinstitute.org

The ClinGen SVI Splicing Subgroup provides recommendations for the application of existing splicing-related ACMG/AMP codes and re-purposing of other codes to capture splicing-related evidence. This study outlines a process for developing a gene-specific PVS1 decision tree and provides methodology to calibrate bioinformatic splice prediction tools.

Brnich et al. *Genome Medicine* (2020) 12:3
<https://doi.org/10.1186/s13073-019-0690-2>

Genome Medicine

GUIDELINE

Open Access

Recommendations for application of the functional evidence PS3/BS3 criterion using the ACMG/AMP sequence variant interpretation framework



Sarah E. Brnich¹ , Ahmad N. Abou Tayoun², Fergus J. Couch³, Garry R. Cutting⁴, Marc S. Greenblatt⁵, Christopher D. Heinen⁶, Dona M. Kanavy¹, Xi Luo⁷, Shannon M. McNulty¹, Lea M. Starita^{8,9}, Sean V. Tavtigian¹⁰, Matt W. Wright¹¹, Steven M. Harrison¹², Leslie G. Biesecker¹³, Jonathan S. Berg^{1*} and On behalf of the Clinical Genome Resource Sequence Variant Interpretation Working Group

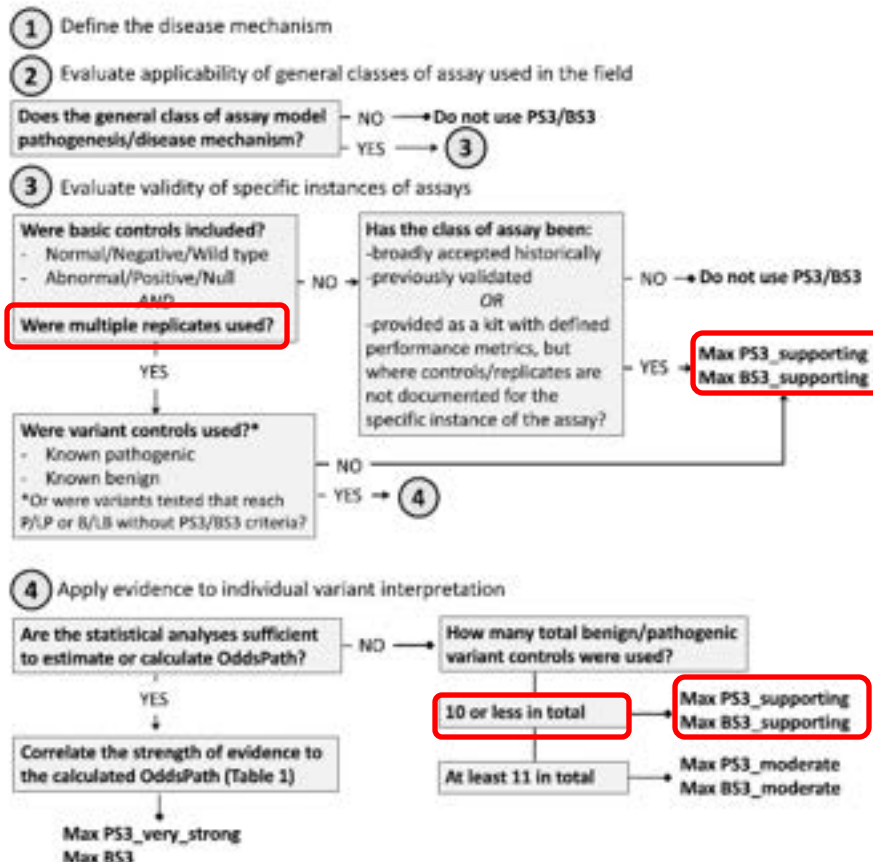


Fig. 2 Decision tree for the evaluation of functional data for clinical variant interpretation. The SM Working Group recommends that evaluators use a four-step process to determine the applicability and strength of evidence of functional assays for use in clinical variant interpretation (evidence codes PS3/BS3): (1) define the disease mechanism, (2) evaluate the applicability of general classes of assay used in the field, (3) evaluate the validity of specific instances of assays, and (4) apply evidence to individual variant interpretation

Use of animal model data in variant classification

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function. To assess the validity of a functional assay, one must consider how closely the functional assay reflects the biological environment. For example, **assaying enzymatic function directly from biopsied tissue from the patient or an animal model provides stronger evidence than expressing the protein in vitro**. Likewise, evidence is stronger if the assay reflects the

ACMG v4.0

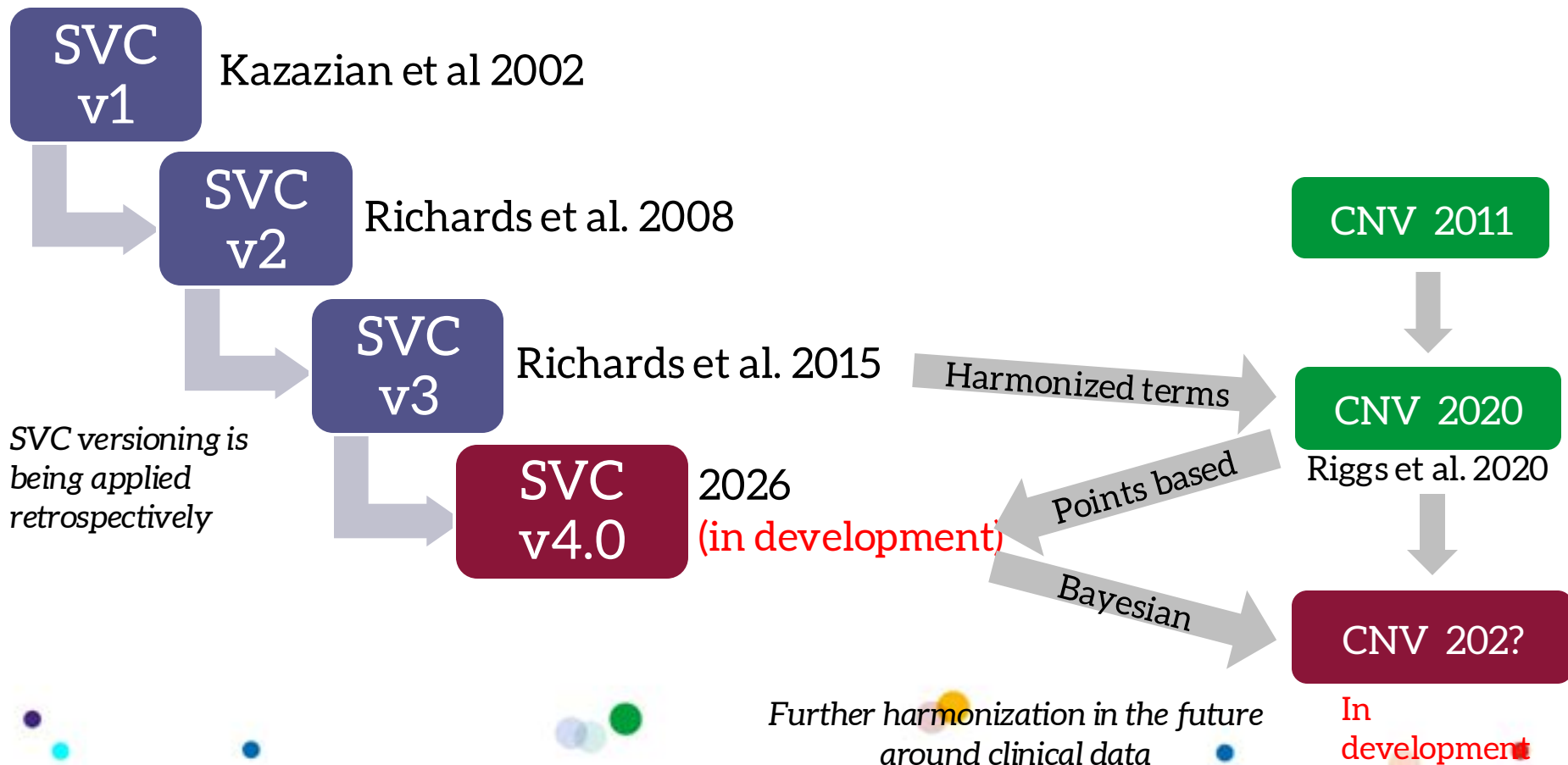
DRAFT

guidelines

Thank you to Steven Harrison
for the following 6 slides



Standards for Sequence Variant and CNV Classification

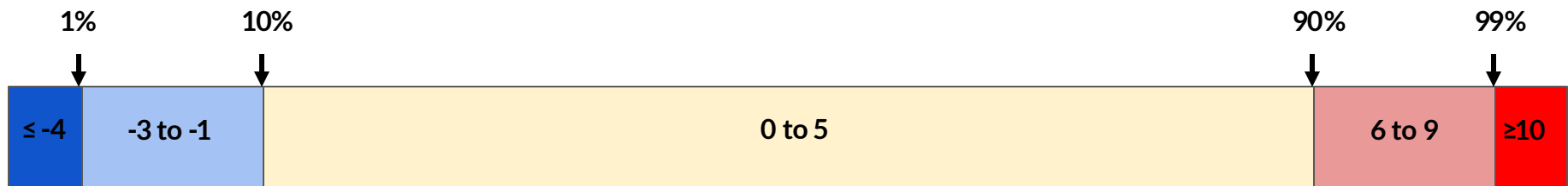


Goals of SVC v4.0

- Revise current standards to address:
 - appropriateness of criteria (PP5/BP6, PM2)
 - ambiguities in the application of criteria
 - strength assigned to criteria, individually and in combination
 - avoidance of double-counting
 - guidance for combining pathogenic and benign criteria
- Incorporate updates from ClinGen working groups, and develop mechanism for ongoing updates
- Where possible, establish more quantitative parameters

	Benign		Pathogenic		
	Strong	Supporting	Supporting	Moderate	Strong
Population Data	MAF is too high for disorder RA2/BS2 OR observation in controls inconsistent with disease penetrance BS2		Absent in population GAB000001 PM2	Prevalence in affecteds statistically increased over controls PS4	
Computational And Predictive Data		Multiple lines of computational evidence suggest no impact BP2 Missense when only truncating cause disease BP2 Silent variant with non-predicted splice impact BP7 In frame indels in repeat without known function BP7	Multiple lines of computational evidence suggest a deleterious effect on the gene/protein product PP3 Novel missense change at an amino acid residue where a different pathogenic missense change has been seen before PM5 Protein length changing variant PM1	Same amino acid change as an established pathogenic variant PS2	Predicted null variant in a gene where LOF is a known mechanism of disease PM1,2
Functional Data	Well-established functional studies show no deleterious effect PS7		Missense in gene with low rate of benign missense variants and path. missense common PP2	Mutation hot spot in well-studied functional domain without benign variation PM3	Well-established functional studies show a deleterious effect PS8
Segregation Data	Non-segregation with disease BS4		Co-segregation with disease in multiple affected family members PP1	Increased segregation data	
De novo Data				De novo (without paternity & maternity confirmed) PM6	De novo (paternity & maternity confirmed) PS2
Allelic Data		Observed in trans with a dominant variant BP2 Observed in cis with a pathogenic variant BP2		For monogenic disorders, detected in trans with a pathogenic variant PM2	
Other (Database)		Reputable source without shared data = benign BNS	Reputable source = pathogenic PPS		
Other Data		Found in case with an alternate cause BP5	Patient's phenotype or not highly specific for gene PM4		

Bayesian 'points' adaptation of SVC v3 = v4 foundation



Benign

Likely
Benign

Uncertain significance

Likely
Pathogenic

Pathogenic

ACMG 2015 Evidence Strength	Odds Path (LR+)	Point Adaption
Benign - Strong	1:18.7	-4
Benign - Supporting	1:2.08	-1
Indeterminate	-	0
Pathogenic - Supporting	2.08:1	+1
Pathogenic - Moderate	4.33:1	+2
Pathogenic - Strong	18.7:1	+4
Pathogenic - Very Strong	350:1	+8

Incorporating Gene-Disease Validity into Variant Classification

<i>Variant Classification</i> <i>G-D validity</i>	PATHOGENIC	LIKELY PATHOGENIC	VUS	LIKELY BENIGN	BENIGN
DEFINITIVE & STRONG	Allowed	Allowed	Allowed	Allowed	Allowed
MODERATE	Not Allowed (limit to LP)	Allowed	Allowed	Allowed	Allowed
LIMITED	Not allowed (limit to VUS)	Not allowed (limit to VUS)	Allowed	Allowed	Allowed
LESS THAN LIMITED (no evidence)	Variant in Gene of Uncertain Significance				
DISPUTED / REFUTED FOR DISEASE	Do not report				

SVC v4 - Hierarchical structure of evidence

Four “root” codes:

- POP_
- CLN_
- LOC_

And then one code from Variant Impact..

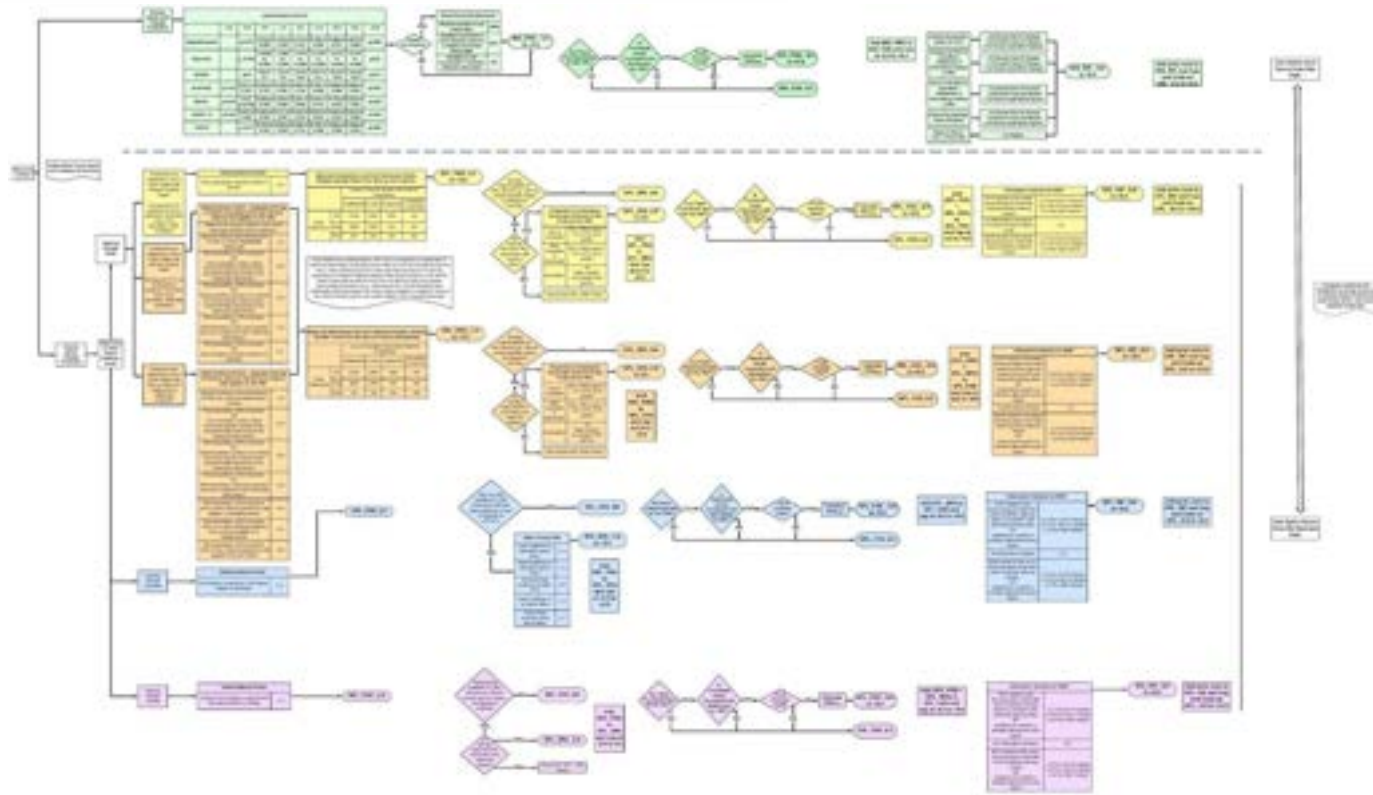
- MIS_
- CDS_
- NUL_
- SPL_

Evidence Category	Evidence Concept	Evidence Code	Code Workflows
Human Observational Data	Population Observations (POP)	POP_FRQ Population Frequency	Allele Frequency
		POP_HMZ Population Homozygotes/Hemizygotes	Hom/Hemi - Autosomal Dominant disorder Hom/Hemi - Autosomal Recessive / X-linked disorder
	Clinical Observations (CLN)	CLN_UAF Unaffected Observations	Unaffecteds - Autosomal Dominant disorder Unaffecteds - Autosomal Recessive / X-linked disorder
		CLN_ALT Alternate Cause	Alternate Cause (Variant) - Autosomal Dominant disorder Alternate Cause (Variant) - Autosomal Recessive / X-linked Alternate Cause (Gene)
		CLN_AFF Affected Observations	Affecteds - Autosomal Dominant disorder Affecteds - Autosomal Recessive / X-linked disorder
		CLN_DNV De Novo Observations	De novo
		CLN_CCS Case-Control Studies	Case - Control
	Locus Specificity (LOC)	LOC_PHE Specific Phenotype	Specific Phenotype
		LOC_SEG Segregations with Disease	Co-segregation - Autosomal Dominant disorder Co-segregation - Autosomal Recessive disorder Co-segregation - Autosomal Semi-Dominant disorder Co-segregation - X-linked disorder
Predictive and Functional Data	Variant Impact (XXX) NB Applicable code determined based on evaluated variant impact	MIS_ Single amino acid change	MIS_PRD (Predicted single amino acid change) MIS_FXN (Functional assessment) MIS_INF (Informative Variants)
		CDS_ Predicted alteration, elongation, or truncation to RNA	CDS_PRD (Predicted Alteration, Elongation or Truncation to RNA) CDS_FXN (Functional assessment) CDS_INF (Informative Variants)
		NUL_ Absent protein	NUL_PRD (Predicted absent protein) NUL_FXN (Predicted absent protein) NUL_INF (Informative Variants)
		SPL_ Alteration to splicing	SPL_PRD (Predicted alteration to splicing) SPL_FXN (Functional assessment) SPL_INF (Informative Variants)

SVC v4 – DRAFT Hierarchical structure of evidence

Evidence Category	Evidence Concept	Evidence Code	Code Workflows	Evidence Code Cap	Evidence Concept Cap	Evidence Category Cap
Human Observational Data	Population Observations (POP)	POP_FRQ Population Frequency	Allele Frequency	-6 - 0	-10 - 0	N/A - N/A
		POP_HMZ Population Homozygotes/Hemizygotes	Hom/Hemi - Autosomal Dominant disorder Hom/Hemi - Autosomal Recessive / X-linked disorder	-4 - 0		
	Clinical Observations (CLN)	CLN_UAF Unaffected Observations	Unaffecteds - Autosomal Dominant disorder Unaffecteds - Autosomal Recessive / X-linked disorder	N/A - 0	N/A - N/A	
		CLN_ALT Alternate Cause	Alternate Cause (Variant) - Autosomal Dominant disorder Alternate Cause (Variant) - Autosomal Recessive / X-linked Alternate Cause (Gene)	N/A - 0		
		CLN_AFF Affected Observations	Affecteds - Autosomal Dominant disorder Affecteds - Autosomal Recessive / X-linked disorder	0 - N/A		
		CLN_DNV De Novo Observations	De novo	0 - +12		
		CLN_CCS Case-Control Studies	Case - Control	-7 - +7		
	Locus Specificity (LOC)	LOC_PHE Specific Phenotype	Specific Phenotype	0 - +4	0 - +4	
		LOC_SEG Segregations with Disease	Co-segregation - Autosomal Dominant disorder	0 - +4		
			Co-segregation - Autosomal Recessive disorder			
			Co-segregation - Autosomal Semi-Dominant disorder			
	Co-segregation - X-linked disorder					
Predictive and Functional Data	Variant Impact (XXX) NB Applicable code determined based on validated variant impact	MIS_ Single amino acid change	MIS_PRD (Predicted single amino acid change)	-4 - +4	-8 - +6	-8 - +9
			MIS_FXN (Functional assessment)	-8 - +8		
			MIS_INF (Informative Variants)	-8 - +8		
		CDS_ Predicted alteration, elongation, or truncation to RNA	CDS_PRD (Predicted Alteration, Elongation or Truncation to RNA)	-4 - +6	-8 - +9	-8 - +10
			CDS_FXN (Functional assessment)	-8 - +8		
			CDS_INF (Informative Variants)	-8 - +8		
		NUL_ Absent protein	NUL_PRD (Predicted absent protein)	-4 - +6	-8 - +10	-8 - +10
			NUL_FXN (Predicted absent protein)	-8 - +8		
			NUL_INF (Informative Variants)	-8 - +8		
		SPL_ Alteration to splicing	SPL_PRD (Predicted alteration to splicing)	-4 - +6	-8 - +8	-8 - +10
			SPL_FXN (Functional assessment)	-8 - +8		
			SPL_INF (Informative Variants)	-8 - +8		

Decision trees = machine readable



THE JOURNEY STARTS HERE



Join the Expedition –
Help Navigate What's Next



Functional data in v4.0 (DRAFT!)

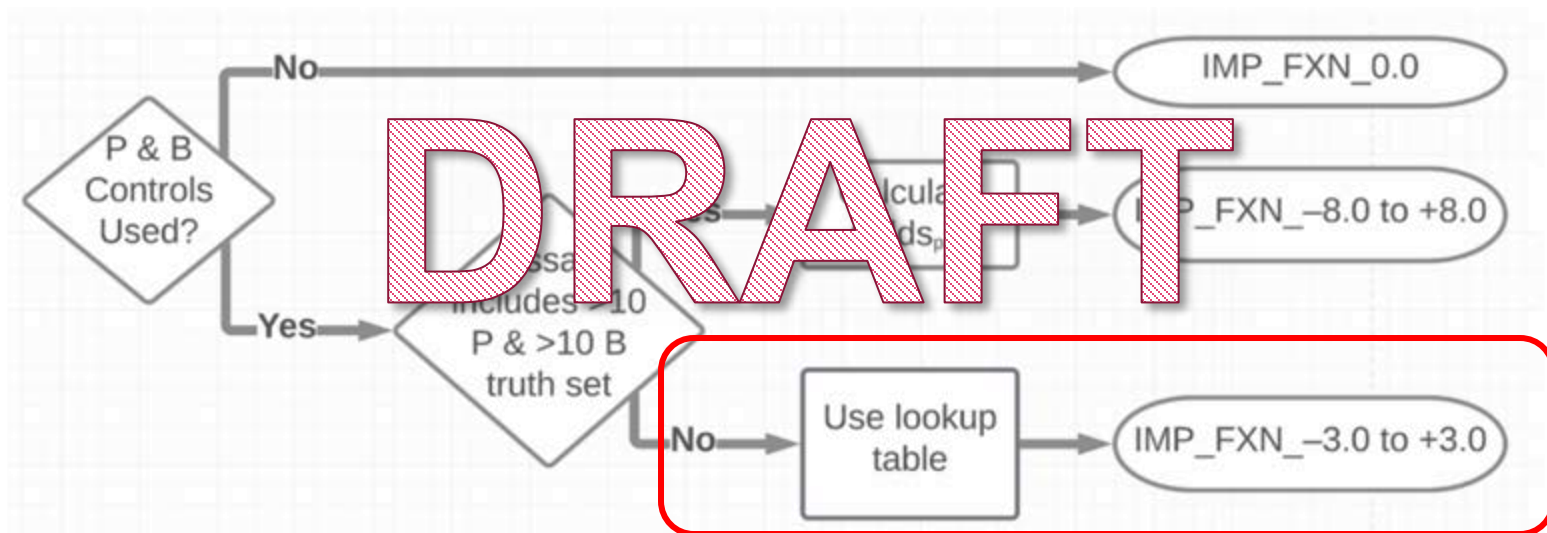


Figure Lingonberry. Flow diagram for evaluation of functional data of protein effect. P: Pathogenic, B: Benign, Odds_{path} is the Bayesian conditional probability, equivalent to a likelihood ratio, as shown by Tavtigian et al (PMID 29300386.)

Genetic Model Organism - ACMG v4.0: FXN_GMO (DRAFT)

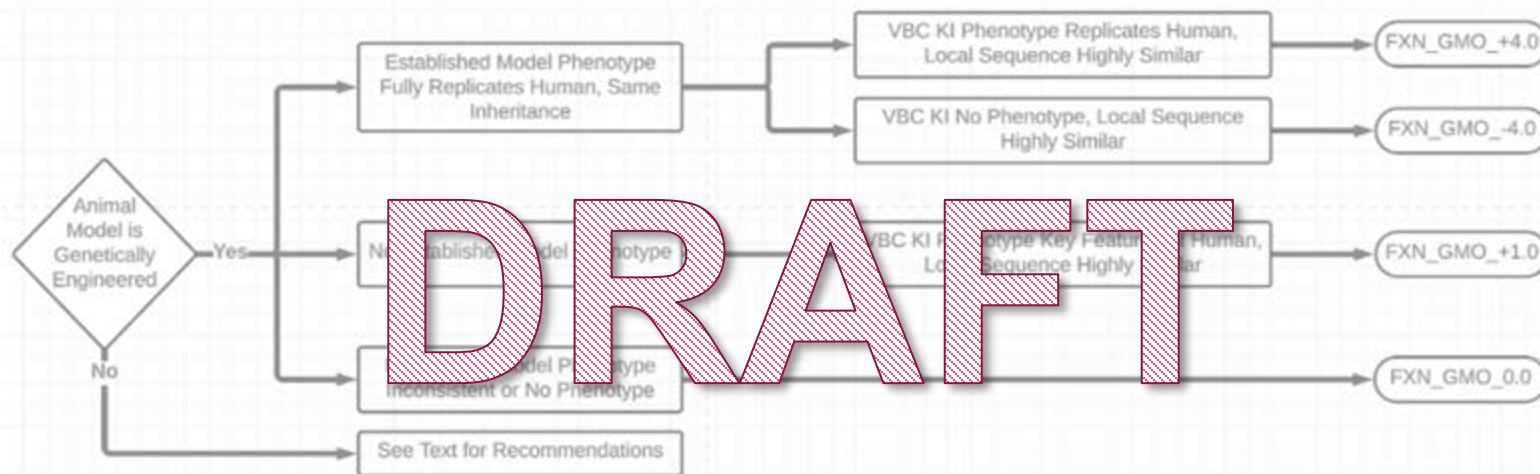


Figure Mulberry. Flow diagram for the use of animal model data based on experiments testing variants for pathogenicity. See text for a discussion of animal models that involve complementation experiments of abnormal genotypes. KI: knock-in

Will variants of uncertain significance still exist in 2030?

Douglas M. Fowler^{1,2,3,*} and Heidi L. Rehm^{4,5,*}



